

S6 Integrated demand-side flexibility

We represent demand-side flexibility as a set of energy-service constraints rather than as an unconstrained reduction in electricity demand. The reference system retains the reconstructed hourly demand profiles, whereas the flexibility case allows selected heating, cooling and electric-vehicle (EV) services to be shifted in time and permits a limited amount of vehicle-to-grid (V2G) discharge. Both cases have identical annual electricity demand, generation and network technologies, security constraints and economic boundaries. Flexible services incur explicit enabling and activation costs, and their contribution to resource adequacy is restricted to the derated power that remains deliverable throughout a four-hour peak window. This formulation allows demand flexibility to reduce operating costs and, where physically justified, firm generation requirements without treating flexible demand as a free or perfectly reliable resource.

S6.1 Demand reconstruction and flexible-service envelopes

Hourly electricity demand is reconstructed for each province and decomposed into residual, space-heating, space-cooling and EV-charging components:

$$L_{p,t}^0 = L_{p,t}^{\text{res}} + L_{p,t}^{\text{heat},0} + L_{p,t}^{\text{cool},0} + L_{p,t}^{\text{EV},0}, \quad (\text{S6-1})$$

where p and t denote province and hour, respectively. The decomposition is closed by construction: heating, cooling and EV charging are internal components of total demand rather than additions to it. Historical hourly profiles are anchored to provincial electricity demand, with the Spring Festival correction modifying intrayear shape while preserving annual electricity consumption. Future residual demand is obtained by subtracting projected heating, cooling and EV electricity from provincial annual demand and distributing the remainder over the retained 8,760-hour residual-load profile.

Heating and cooling loads are derived from population- and electricity- weighted ERA5 meteorology [1]. Temperature, solar irradiance, wind speed and humidity are combined in a building-adjusted temperature index with a 48-hour exponentially weighted memory, after which region-specific heating- and cooling-degree thresholds are applied. For each province and hour, the response of the reconstructed thermal load to a $\pm 1^\circ\text{C}$ comfort perturbation defines the maximum upward and downward modulation. These weather- and load-dependent envelopes prevent the optimization from creating thermal storage independently of the underlying end use. The use of an equivalent energy state to represent aggregated building flexibility is consistent with established characterizations of thermal mass and controlled heating, ventilation and air-conditioning loads [2, 3, 4, 5, 6].

The exogenous EV-charging profile is constructed from provincial new-energy vehicle stocks, monthly electricity use per vehicle and normalized intraday charging probabilities. For planning year y ,

$$L_{p,h,y}^{\text{EV},0} = \frac{N_{p,y}^{\text{EV}} e_{m(h)} \pi_h}{1000}, \quad (\text{S6-2})$$

where $N_{p,y}^{\text{EV}}$ is the provincial vehicle stock, $e_{m(h)}$ is daily charging electricity per vehicle in month m , and π_h is the normalized hourly charging probability. Coordinated charging

does not alter vehicle stocks, annual mobility energy or the uncontrolled charging reference. Instead, a prescribed share of this reference is made schedulable within power, energy and departure-service limits.

The central case enrolls 25% of heating demand, 20% of cooling demand and 15% of EV charging in flexible-service contracts. These fractions scale the physical envelopes rather than the underlying demand. They are treated as scenario parameters because national demand-response measures and local air-conditioning programmes establish the institutional feasibility of contracted flexibility, but do not provide nationally representative participation rates [7, 8, 9, 10]. Observed responses in Chinese coordinated-charging programmes similarly support a non-zero schedulable EV share while revealing substantial heterogeneity in participation and price response [14, 15].

S6.2 Joint thermal-load and electric-vehicle flexibility model

For thermal service $s \in \{\text{heat}, \text{cool}\}$, the optimization chooses contracted capacity K_p^s , upward modulation $P_{p,t}^{s,+}$ and downward modulation $P_{p,t}^{s,-}$. The hourly decisions are bounded by the comfort-derived envelopes, $\bar{P}_{p,t}^{s,+}$ and $\bar{P}_{p,t}^{s,-}$, and by the contracted power available at that hour, $a_{p,t}^s K_p^s$:

$$0 \leq P_{p,t}^{s,+} \leq \min\left\{\bar{P}_{p,t}^{s,+}, a_{p,t}^s K_p^s\right\}, \quad (\text{S6-3})$$

$$0 \leq P_{p,t}^{s,-} \leq \min\left\{\bar{P}_{p,t}^{s,-}, a_{p,t}^s K_p^s\right\}. \quad (\text{S6-4})$$

An equivalent thermal-service state $S_{p,t}^s$ tracks the intertemporal consequence of pre-heating or pre-cooling:

$$S_{p,t}^s = \rho_p^s S_{p,t-1}^s + \eta_p^{s,+} P_{p,t}^{s,+} - \frac{P_{p,t}^{s,-}}{\eta_p^{s,-}}, \quad 0 \leq S_{p,t}^s \leq d_p^s K_p^s. \quad (\text{S6-5})$$

Here, ρ_p^s is hourly service retention and d_p^s is the maximum energy-to-power duration. The state is non-negative and periodic over the modelled year, precluding terminal comfort debt. The resulting electrical load is $\tilde{L}_{p,t}^s = L_{p,t}^{s,0} + P_{p,t}^{s,+} - P_{p,t}^{s,-}$. Central hourly retention is 0.94 for heating and 0.92 for cooling, with corresponding service durations of 4 and 5 hours.

EV flexibility is represented by a single aggregate mobility-service state rather than by superimposing a second battery on coordinated charging. A fraction α^{V1G} of the uncontrolled EV load is schedulable, whereas the remainder is fixed:

$$L_{p,t}^{\text{EV},\text{flex},0} = \alpha^{\text{V1G}} L_{p,t}^{\text{EV},0}, \quad L_{p,t}^{\text{EV},\text{fixed}} = (1 - \alpha^{\text{V1G}}) L_{p,t}^{\text{EV},0}. \quad (\text{S6-6})$$

Let $P_{p,t}^{\text{ch}}$ and $P_{p,t}^{\text{out}}$ denote flexible charging and V2G discharge, respectively. The fleet-service state evolves as

$$E_{p,t}^{\text{EV}} = E_{p,t-1}^{\text{EV}} + \eta^{\text{ch}} P_{p,t}^{\text{ch}} - D_{p,t}^{\text{mob}} - \frac{P_{p,t}^{\text{out}}}{\eta^{\text{dis}}}, \quad (\text{S6-7})$$

where $D_{p,t}^{\text{mob}}$ is exogenous driving-energy withdrawal. The state is bounded by the hourly fleet-energy envelope and by a minimum departure-energy requirement, and it is periodic

over the modelled year. Charging and discharge are further constrained by exogenous hourly power envelopes and endogenous contracted capacities:

$$0 \leq P_{p,t}^{\text{ch}} \leq \min \left\{ \bar{P}_{p,t}^{\text{ch}}, a_{p,t}^{\text{EV}} K_p^{\text{V1G}} \right\}, \quad (\text{S6-8})$$

$$0 \leq P_{p,t}^{\text{out}} \leq \min \left\{ \bar{P}_{p,t}^{\text{out}}, a_{p,t}^{\text{EV}} K_p^{\text{V2G}} \right\}, \quad (\text{S6-9})$$

$$0 \leq K_p^{\text{V2G}} \leq K_p^{\text{V1G}}. \quad (\text{S6-10})$$

Thus, bidirectional participation is nested within the coordinated-charging contract. It cannot exceed either the participating fleet or its hourly power and energy availability. Charge and discharge efficiencies are both 0.94 in the central case.

The national sum of contracted V2G power is capped at 10, 20, 30 and 40 GW in 2030, 2040, 2050 and 2060, respectively. The 2030 value is anchored to China's policy objective of establishing bidirectional interaction at the ten-gigawatt scale, and the subsequent values are explicit planning extrapolations rather than forecasts. The national vehicle-grid policy and the first large-scale pilot programme, covering nine cities and 30 projects, provide the deployment context [12, 13]. The aggregate representation deliberately retains conservative service constraints because city-scale evidence shows that mobility, dwell time, connection, compensation, range requirements and battery degradation jointly determine usable V2G potential [16, 17].

S6.3 System integration, economic accounting and capacity adequacy

The hourly load entering the power balance is

$$\begin{aligned} L_{p,t}^{\text{eff}} = & L_{p,t}^{\text{res}} + \tilde{L}_{p,t}^{\text{heat}} + \tilde{L}_{p,t}^{\text{cool}} \\ & + L_{p,t}^{\text{EV, fixed}} + P_{p,t}^{\text{ch}} - P_{p,t}^{\text{out}}. \end{aligned} \quad (\text{S6-11})$$

This effective load is used consistently in hourly power balance and in load-dependent operating-security requirements. Flexible demand receives no additional reserve credit, which avoids using the same deliverable power simultaneously for resource adequacy and operating reserves.

All flexible services incur real-resource costs. Thermal flexibility pays an annual enabling cost and an activation cost on both upward and downward modulation. Coordinated EV charging pays an annual control and aggregation cost and an inconvenience cost on charging energy displaced from its uncontrolled hour. To prevent V2G replenishment from being misclassified as coordinated-charging activity, relocated energy is defined by

$$R_{p,t}^{\text{V1G}} \geq L_{p,t}^{\text{EV, flex, 0}} - P_{p,t}^{\text{ch}}, \quad R_{p,t}^{\text{V1G}} \geq 0. \quad (\text{S6-12})$$

V2G additionally pays for availability, annualized bidirectional charging infrastructure, owner participation per unit of discharge and battery degradation per unit of discharge.

The incremental flexibility cost is

$$\begin{aligned}
C^{\text{flex}} = & \sum_{p,s} c_s^{\text{ena}} K_p^s + 10^{-3} \sum_{p,t,s \in \{\text{heat}, \text{cool}\}} c_s^{\text{act}} (P_{p,t}^{s,+} + P_{p,t}^{s,-}) \\
& + 10^{-3} \sum_{p,t} c_{\text{V1G}}^{\text{reloc}} R_{p,t}^{\text{V1G}} + \sum_p c_{\text{V2G}}^{\text{infra}} K_p^{\text{V2G}} \\
& + 10^{-3} \sum_{p,t} \left(c_{\text{V2G}}^{\text{owner}} c_{\text{V2G}}^{\text{deg}} \right) P_{p,t}^{\text{out}}.
\end{aligned} \tag{S6-13}$$

Costs are expressed in 2025 constant CNY. Time-of-use tariffs and public subsidies are treated as transfers and are not added to social system cost; metering, control, aggregation, infrastructure, inconvenience and battery wear are treated as real resource costs. Charging electricity and conversion losses remain endogenous in the system energy balance and are therefore not charged a second time. This accounting follows the broader evidence that the system value of demand flexibility depends on non-zero enabling, interoperability and participation costs [11, 18, 19].

Resource adequacy is assessed against the immutable annual provincial peak, rather than against an endogenous peak that could be reduced without a deliverability test:

$$C_p^{\text{credit}} + \sum_s F_p^{s,\text{firm}} \geq (1 + m) \max_t L_{p,t}^0, \tag{S6-14}$$

where C_p^{credit} is supply-side accredited capacity, m is the planning margin and $F_p^{s,\text{firm}}$ is the accredited contribution of flexible service s . For each province, the deliverability window comprises the annual peak hour, the preceding hour and the following two hours. Thermal and coordinated-charging credits are bounded by contracted power and the minimum reduction envelope throughout this four-hour window. V2G credit is additionally bounded by the minimum fleet energy in the window divided by four hours:

$$F_p^{s,\text{firm}} \leq \delta_s \min \left\{ K_p^s, \min_{t \in \mathcal{W}_p} \bar{P}_{p,t}^{s,-} \right\}, \quad s \in \{\text{heat}, \text{cool}\}, \tag{S6-15}$$

$$F_p^{\text{V1G},\text{firm}} \leq \delta_{\text{V1G}} \min \left\{ K_p^{\text{V1G}}, \min_{t \in \mathcal{W}_p} L_{p,t}^{\text{EV},\text{flex},0} \right\}, \tag{S6-16}$$

$$F_p^{\text{V2G},\text{firm}} \leq \delta_{\text{V2G}} \min \left\{ K_p^{\text{V2G}}, \min_{t \in \mathcal{W}_p} \bar{P}_{p,t}^{\text{out}}, \frac{\min_{t \in \mathcal{W}_p} \bar{E}_{p,t}^{\text{fleet}}}{4} \right\}. \tag{S6-17}$$

Central derating factors are 0.60 for heating and cooling and 0.50 for coordinated charging and V2G. This approach preserves the capacity value of demand flexibility while preventing a one-hour reduction in effective load from being interpreted as perfectly firm capacity. It is consistent with the principle that capacity accreditation should reflect sustained, system-specific deliverability [20, 21, 22]. Demand response and storage capacity value have also been incorporated in Chinese expansion-planning research [23]; however, the adopted derating factors are engineering assumptions, not empirically estimated provincial ELCC values. Regulatory ELCC treatments are used only as methodological comparators [24].

S6.4 Parameter evidence, sensitivities and interpretation

Table S6-1 summarizes the central assumptions and the prespecified ranges used to evaluate structural uncertainty. The parameter set combines reproducible end-use envelopes with Chinese policy and behavioural evidence, peer-reviewed technology studies and international resource-adequacy guidance. Policy targets and compensation rules are used to establish feasibility and scale, rather than being transferred directly into national social-cost parameters.

Table S6-1: Central assumptions and prespecified sensitivity ranges for integrated demand-side flexibility. Monetary values are in 2025 constant CNY.

Parameter	Central value	Sensitivity range	Evidence and interpretation
Heating and cooling enrollment	0.25; 0.20	0.10–0.40	Applied to weather- and comfort-derived envelopes. Chinese demand-response rules and local air-conditioning programmes establish feasibility but not a national participation rate [7, 8, 9, 10].
Heating and cooling hourly retention	0.94; 0.92	0.90–0.98; 0.85–0.97	Equivalent-service representation informed by building thermal-flexibility studies [3, 5, 6].
Heating and cooling duration	4 h; 5 h	2–8 h; 2–10 h	Energy-to-power limit for thermal service, evaluated across the range reported for controlled HVAC and thermal mass [2, 4].
Thermal enabling cost	40 CNY/kW-y	10–120 CNY/kW-y	Control, metering, aggregation and availability; treated as a real-resource cost [11, 7].
Thermal activation cost	300 CNY/MWh	100–1000 CNY/MWh	Service activation and comfort cost. Local compensation schedules provide scale comparators but are not added as fiscal transfers [9, 10].
Coordinated-charging enrollment	0.15	0.10–0.25	Share of uncontrolled EV charging that can be rescheduled; informed by Chinese pilot participation and price-response evidence [14, 15].

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Table S6-1 continued

Parameter	Central value	Sensitivity range	Evidence and interpretation
V2G participation envelope	0.10	0.05–0.30	Scales the provincial discharge envelope; does not represent an hourly plug-in probability [12, 17].
Charging and discharge efficiencies	0.94; 0.94	0.90–0.97	Grid-to-fleet and fleet-to-grid conversion efficiencies [16, 17].
Coordinated-charging power ratio and service duration	2.0; 24 h	1.5–3.0; 12–36 h	Aggregate power and energy limits evaluated as scenario uncertainty; the service state is not interpreted as nameplate vehicle-battery capacity.
Coordinated-charging enabling and relocation costs	60 CNY/kW-y; 300 CNY/MWh	20–180 CNY/kW-y; 100–400 CNY/MWh	Control and aggregation cost plus participant inconvenience on charging energy displaced once [14, 15].
V2G availability and infrastructure costs	60; 40 CNY/kW-y	20–180; 20–100 CNY/kW-y	Incremental availability and annualized bidirectional equipment/platform costs [16, 18].
V2G owner and degradation costs	150; 400 CNY/MWh	100–300; 250–500 CNY/MWh	Participation compensation and battery wear are represented separately [18, 19].
Thermal firm-capacity derating	0.60	0.30–0.80	Applied only to four-hour deliverable reduction; not a China-specific ELCC estimate [20, 21, 22].
Coordinated-charging and V2G firm-capacity derating	0.50; 0.50	0.20–0.70	Applied to four-hour power and energy deliverability; not a China-specific ELCC estimate [23, 24].
National V2G contracted-power ceiling	10/20/30/40 GW	5–10/10–30/15– 50/20–70 GW	2030 is anchored to China’s ten-gigawatt-scale objective; later values are planning extrapolations [12, 13].

The integrated formulation is intended to identify mechanisms rather than to assign a deterministic forecast to demand-side participation. Its results should therefore be interpreted jointly with the low and high assumptions in Table S6-1. In particular, a reduction in generation capacity is attributable to demand-side flexibility only when the associated four-hour firm-credit constraint is binding and the full enabling, activation, infrastructure and degradation costs are included. A reduction in the unconstrained effective peak alone is not sufficient evidence of firm capacity substitution.

Several limitations remain. Building flexibility is represented by province-level comfort envelopes rather than building-by-building thermal models. EV mobility and connection are aggregated, and the model does not resolve individual trip chains, charger locations or distribution-network constraints. The four-hour accreditation window is a transparent deterministic approximation and does not replace a probabilistic loss-of-load or ELCC assessment. Finally, periodic service states are appropriate for an annual planning model but should not be interpreted as evidence about behaviour beyond the represented horizon. These limitations motivate sensitivity analysis of participation, duration, costs and capacity derating, and constrain scientific interpretation to complete annual simulations.

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